

Solving problems with the help of equations

Name the unknown, write the sentence as an equation, solve it, and answer the question that was asked.

LESSON 127–128

2 × 40 MIN

MATEMATIKA 7, §57

S8 2.5

LESSON CLOCK

10'

25'

25'

20'

5'

The five steps	10 min
Choosing the unknown	25 min
Three families of problem	25 min
Checking against the words	20 min
Homework	5 min

LEARNING OBJECTIVES

- Choose the unknown so that the other quantities are easy to express.
- Translate a worded condition into an equation.
- Solve the equation and check the root against the original words.
- Reject a root that is impossible in the situation described.

TEXTBOOK

National	pp. 378–386
Cambridge	Stage 8, pp. 31–36
Workbook	Exercise 2.6

01

Explanation

The five steps

STEP	WHAT TO DO
1	let x be one of the unknown quantities, and say in words which
2	write every other quantity in terms of x
3	find the sentence in the problem that is an equality, and write it as an equation
4	solve the equation
5	answer the question that was asked, in words and units

STEP 5 IS NOT AUTOMATIC

If x is the son's age but the question asks for the father's, the root is not the answer. Read the question again after solving — this is where whole marks are lost for correct algebra.

Choosing the unknown

Let x be the quantity that makes the others simplest to write — usually the smallest one, or the one the others are compared with.

PROBLEM	LET x BE	THEN
a father is 3 times as old as his son	the son's age	the father is $3x$
two numbers differ by 7	the smaller	the larger is $x + 7$
a rectangle is 4 <i>cm</i> longer than it is wide	the width	the length is $x + 4$
a sum of 120 split in the ratio 2 : 3	one part of the ratio	the shares are $2x$ and $3x$

A BAD CHOICE IS NOT WRONG, ONLY SLOWER

Taking the father's age as x makes the son $\frac{x}{3}$ and brings fractions into the equation. The answer is the same; the work is not.

Three families of problem

FAMILY	PROBLEM	EQUATION	ANSWER
ages	a father is 3 times as old as his son; in 12 years he will be twice as old	$3x + 12 = 2(x + 12)$	son 12, father 36
numbers	two numbers differ by 7 and add to 39	$x + (x + 7) = 39$	16
geometry	a rectangle is 4 <i>cm</i> longer than wide, perimeter 36 <i>cm</i>	$2(x + x + 4) = 36$	7 <i>cm</i>
sharing	120 split in the ratio 2 : 3	$2x + 3x = 120$	48

FIND THE EQUALITY SENTENCE

Every problem contains exactly one sentence that says two things are equal — “add to”, “will be twice”, “the perimeter is”. That sentence, and only that one, becomes the equation.

Checking against the words

The check is against the problem, not against the equation. And a root that cannot be true of the situation must be rejected.

ROOT	QUESTION	VERDICT
$x = 12$	the son's age	accepted
$x = -5$	a length	rejected — a length is positive
$x = 7.5$	a number of pupils	rejected — it must be whole
$x = 0$	a speed	usually rejected — nothing moves

REJECTING A ROOT IS PART OF THE ANSWER, NOT A FAILURE

An equation knows nothing about ages and lengths. Deciding which of its roots the situation allows is the mathematician's job, and saying so explicitly earns the mark.

02

Worked examples

EXAMPLE 1 A father is 3 times as old as his son. In 12 years he will be twice as old. How old is each now?

1

Let x be the son's age; the father is $3x$.

2

In 12 years: son $x + 12$, father $3x + 12$.

3

$3x + 12 = 2(x + 12)$
The equality sentence.

4

$3x + 12 = 2x + 24$, so $x = 12$.
Son 12, father 36.

ANSWER

Son 12, father 36

EXAMPLE 2 The perimeter of a rectangle is 36 cm and it is 4 cm longer than it is wide. Find its sides.

- ① Let the width be x ; the length is $x + 4$.
- ② $2(x + x + 4) = 36$
- ③ $4x + 8 = 36$, so $x = 7$.
- ④ Width 7 cm, length 11 cm. Check: $2(7 + 11) = 36$ ✓

ANSWER

7 cm by 11 cm

EXAMPLE 3 Two numbers differ by 7 and add to 39. Find them.

- ① Let the smaller be x ; the larger is $x + 7$.
- ② $x + (x + 7) = 39$
- ③ $2x = 32$, so $x = 16$.
- ④ The numbers are 16 and 23.

ANSWER

16 and 23

03 Interactive model

Ask the class to write only step 1 and step 3 for six problems, without solving any of them; translation is the skill, and separating it from the arithmetic doubles the practice.

Translate the sentence

“A number increased by 5” is:

$$5x$$

$$x + 5$$

$$x - 5$$

$$\frac{x}{5}$$

“Three times a number, less 2” is:

$$3x - 2$$

$$3(x - 2)$$

$$2 - 3x$$

$$x - 6$$

“Three times a number less 2, all doubled” is:

$$2 \cdot 3x - 2$$

$$2(3x - 2)$$

$$6x - 4 \text{ only}$$

$$3x - 4$$

If the son is x and the father is 3 times as old, the father is:

$$x + 3$$

$$3x$$

$$\frac{x}{3}$$

$$x - 3$$

In 12 years the father will be:

$$3x$$

$$3x + 12$$

$$3(x + 12)$$

$$3x - 12$$

The equation is then:

$$3x + 12 = 2(x + 12)$$

$$3x = 2x + 12$$

$$3x + 12 = 2x$$

$$3(x + 12) = 2x$$

A rectangle 4 longer than wide has perimeter:

$$2x + 4$$

$$4x + 8$$

$$x(x + 4)$$

$$4x + 4$$

A root of -5 for a length is:

accepted

rejected

doubled

squared

04 Terminology

Write these on the board at the start. Learners meet the national programme in Uzbek or Russian and the Cambridge papers in English — the three columns are the same idea.

ENGLISH	O'ZBEKCHA	РУССКИЙ
Word problem	Matnli masala	Текстовая задача
To denote	Belgilamoq	Обозначить
Condition	Shart	Условие
To translate	O'girmoq	Перевести
Total	Jami	Всего
Speed	Tezlik	Скорость
Impossible answer	Mos kelmaydigan javob	Посторонний ответ
To interpret	Talqin qilmoq	Истолковать

Stress the words in bold in the explanation above; ask for the Uzbek and Russian back before moving on.

05 Quick check**Check yourself**

The first step is:

writing the equation

naming the unknown

solving

checking

The equation comes from:

the question asked

the equality in the problem

the units

the answer

If the son is x and the father $3x$, in 12 years the father is:

$3x$

$3x + 12$

$3(x + 12)$

$3x + 36$

A rectangle 4 *cm* longer than wide with perimeter 36 has width:

Two numbers differ by 7 and add to 39: the smaller is:

A root of 7.5 pupils is:

06 Practice · 21 problems

Seven at each level. Set the easy row for everyone, the medium row for the main body of the class, and the hard row for those who finish early.

Easy · warm the idea up

7 PROBLEMS

1. "A number increased by 5"

ANSWER $x + 5$

2. "Three times a number, less 2"

ANSWER $3x - 2$

3. "Half a number"

ANSWER $\frac{x}{2}$

4. A father is 3 times a son aged x

ANSWER $3x$

5. Two numbers differing by 7, smaller x

ANSWER x and $x + 7$

6. A rectangle of width x and length $x + 4$: its perimeter

ANSWER $4x + 8$

7. 120 in the ratio 2 : 3

ANSWER $2x$ and $3x$

Medium · the standard question

7 PROBLEMS

1. Two numbers differ by 7 and add to 39

ANSWER 16 and 23

2. A rectangle 4 cm longer than wide, perimeter 36 cm

ANSWER 7 cm by 11 cm

3. 120 split in the ratio 2 : 3

ANSWER 48 and 72

4. A father 3 times his son; in 12 years twice as old

ANSWER Son 12, father 36

5. A number is 5 more than another and they total 31

ANSWER 13 and 18

6. Three consecutive numbers add to 72

ANSWER 23, 24, 25

7. A pencil costs x and a pen $3x$; together 8000 sum

ANSWER Pencil 2000, pen 6000

Hard · stretch and reason

7 PROBLEMS

1. A mother is 4 times her daughter; in 20 years twice as old

ANSWER Daughter 10, mother 40

2. The perimeter of a rectangle is 44 and the length is 3 times the width

ANSWER 5.5 by 16.5

3. Three consecutive even numbers add to 90

ANSWER 28, 30, 32

4. A sum of 300 is split so that the second share is 20 more than the first and the third twice the first

ANSWER 70, 90, 140

5. Half a number less 3 equals a third of it plus 1

ANSWER $x = 24$

6. A father was 5 times his son 4 years ago and is 3 times now

ANSWER Son 8, father 24

7. Why must a root of -5 be rejected for a length?

ANSWER A length cannot be negative

07 Homework

Homework — 5 tasks

Write “Let x be ...” in words before any algebra, and answer in words at the end.

- Two numbers differ by 9 and add to 45. Find them.
- A rectangle is 5 cm longer than it is wide and has perimeter 42 cm. Find its sides.
- A mother is 4 times as old as her daughter; in 20 years she will be twice as old. Find both ages.
- Split 210 in the ratio 3 : 4.

5. Three consecutive numbers add to 81. Find them.